

Bank Customer Churn Prediction Using Random Forest and XGBoost Machine Learning Algorithms

Machine Learning

BACHELOR OF TECHNOLOGY

in

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DEPARTMENT OF _____

Name of Title: **Bank Customer Churn Prediction Using Random Forest and XGBoost Machine Learning Algorithms**

1. Abstract:

Customer churn is an important problem in the banking sector because losing existing customers can negatively affect revenue and business growth. This project focuses on developing a machine learning-based customer churn prediction system that can identify customers who are likely to leave a bank. The system uses customer-related information such as age, gender, geography, credit score, tenure, account balance, number of products, credit card status, active membership, and estimated salary. A dataset containing customer records is first preprocessed by handling missing values and converting categorical data into a suitable numerical format for machine learning algorithms. Relevant features are selected to improve the prediction process. Two machine learning algorithms, Random Forest and XGBoost, are used for classification. Random Forest combines multiple decision trees to provide reliable predictions and identify important features. XGBoost uses gradient boosting to build an efficient and accurate prediction model by improving errors from previous trees. The dataset is divided into training and testing sets, and both models are trained using the training data. Their performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The results of both algorithms are compared to determine which model performs better for customer churn prediction. The proposed system can help banks identify high-risk customers at an early stage and take suitable retention measures such as personalized offers, improved services, and customer engagement strategies. This can help reduce customer loss, improve customer satisfaction, and support better decision-making in the banking sector.

2. INTRODUCTION:

Customer churn means when a customer stops using a bank's services. Predicting churn helps banks identify customers who may leave and take action to retain them. This project uses machine learning to predict customer churn based on information such as age, credit score, balance, tenure, and account details. Random Forest and XGBoost algorithms are used to train the models, and their performance is compared using different evaluation metrics. The system helps banks identify high-risk customers and improve customer retention.

3. SOFTWARE REQUIREMENTS:

- Python – Programming
- Pandas & NumPy – Data processing
- Scikit-learn – Random Forest and evaluation
- XGBoost – XGBoost model
- Google Colab/Jupyter Notebook – Running the project
- Matplotlib & Seaborn – Data visualization

Signature of the Guide

Course Instructor